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OBSERVATION / DATA SHEET

Date 19.12.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 10

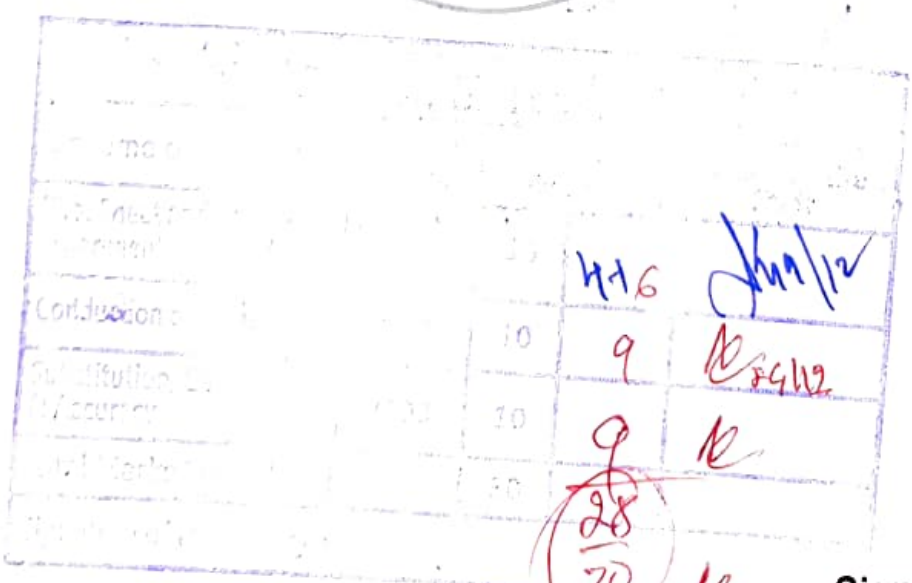
Title Numerical aperture and attenuation coefficient of an optical fiber.

Aim: Part A: To determine the Numerical aperture of the given optical fiber

Part B: To measure the attenuation coefficient of the given optical fiber.

Apparatus: Optical Fiber Kit, Optical fibre Cables, In-line adapter Numerical Aperture Jig.

PART A: To determine the Numerical aperture of Optical fiber



		4+6	
Conclusion	10	9	
	10	9	
		28/30	

07/01/24

Signature of Teacher Incharge

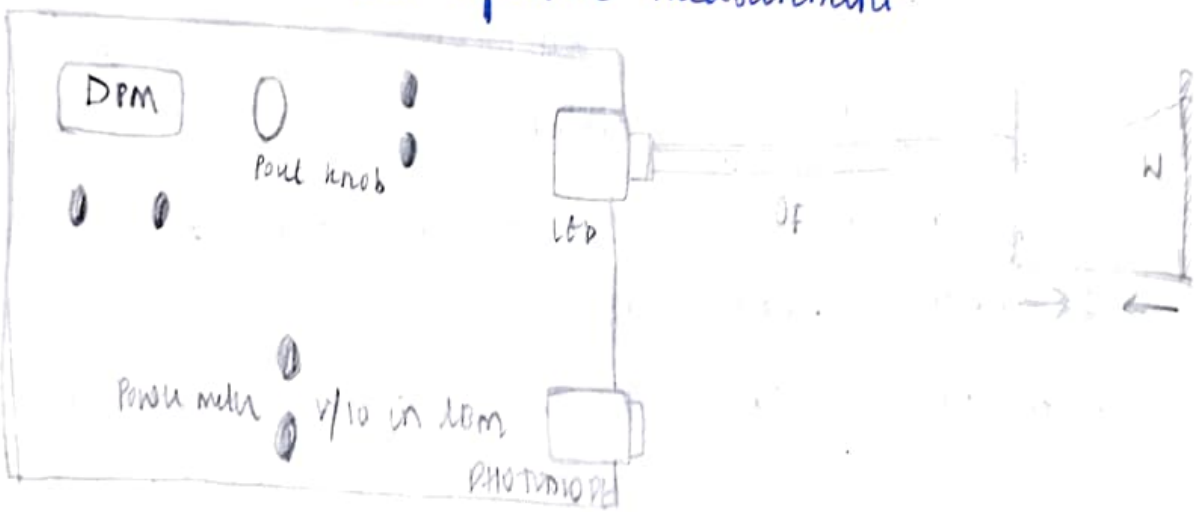
Formula : $NA = \sin\theta_0 = \frac{W}{\sqrt{(4L^2 + W^2)}}$

W → diameter of the beam spot, L → distance from the Opt
fibre to the screen

OBSERVATIONS:

Diagram : Experimental Setup :

Part A : Numerical aperture measurement



Numerical Aperture $N.A = \sin\theta_0 = \frac{W}{\sqrt{(4L^2 + W^2)}}$

where W → diameter of the beam spot, L → distance fro
Optical fiber to th

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Date 19 12 23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 10

Title Numerical aperture and attenuation coefficient of an optical fiber

PART B :



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TABLE A :

Sl No	L (mm)	W_1 (mm)	W_2 (mm)	$W = (W_1 + W_2)/2$ (mm)	Numerical aperture	Acceptance angle $\theta = \sin^{-1}(NA)$ in degree
			14/9/12			
1.	2	12	12	12	0.949	71.62

high!

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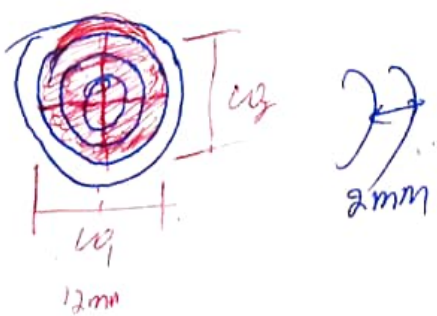
Sl no	L (mm)	W_1 (mm)	W_2 (mm)	$W = (W_1 + W_2/2)$ (mm	Numerical aperture (NA)	Acceptance angle $\theta = \sin^{-1}(NA)$ in degree
2.	4	16	16	16	0.894	63.38
3.	6	20	20	20	0.857	58.98
4.	8	24	24	24.	0.700	44.42

Table B

Length (m)	(A) Attenuation in dB	Length (m)	(B) attenuation in dB	(B-A) Attenuation for 4m length i
1.	-11.23	5.	-11.76.	-0.53

Attenuation coefficient : Attenuation per unit length (α)

$$\alpha = \frac{\text{Attenuation loss}}{\text{length}} = \frac{-0.53}{4} = -0.1325 \text{ dB/m} = \frac{-0.1325 \times}{} \text{ dB}$$



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Date 19 12 23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 10

Title Numerical Aperture

Calculations: $NA = \sin\theta_0 = \frac{W}{\sqrt{4L^2+W^2}}$

$NA = \frac{12}{\sqrt{4(2)^2+12^2}} = \frac{12}{\sqrt{160}} = \frac{12}{12.64} = 0.949$

$\theta = \sin^{-1}(NA)$

$= \sin^{-1}(0.949) = 71.62.$

$NA = \frac{16}{\sqrt{4(4)^2+16^2}} = \frac{16}{\sqrt{4(16)+256}} = \frac{16}{\sqrt{320}} = \frac{16}{17.88} = 0.894$

$\theta = \sin^{-1}(0.894) = 63.38$

$NA = \frac{20}{\sqrt{4(6)^2+20^2}} = \frac{20}{\sqrt{4(36+400}} = \frac{20}{\sqrt{544}} = \frac{20}{23.32} = 0.857$

$\theta = \sin^{-1}(0.857) = 58.98$

$NA = \frac{24}{\sqrt{4(8)^2+24^2}} = \frac{24}{\sqrt{4(64)+576}} = \frac{24}{\sqrt{832}} = \frac{20}{28.84}$

$= 0.700$

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$$\theta = \sin^{-1}(0.700) = 44.42.$$

$$NA = \frac{0.949 + 0.894 + 0.857 + 0.700}{4} = \frac{3.4}{4} = 0.85$$

$$\theta = \frac{71.62 + 63.38 + 58.98 + 44.42}{4} = \frac{238.4}{4} = 59.6^{\circ}$$

Result :

1.	the numerical aperture of the given optical fiber	0.85
2.	the acceptance angle θ is	59.6
3.	the attenuation coefficient of the fibre α	-0.1345×10^{5}

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OBSERVATION / DATA SHEET

Date 12.12.23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 8

Title DIELECTRIC CONSTANT

Aim: To determine the capacity of a parallel plate capacitor and hence to calculate the dielectric constant of the dielectric medium in it.

Apparatus: Battery of ten volts, electrolytic capacitor, digital multimeter, two way key and stop clock.

Formula: The capacitance and dielectric constant of the given capacitor are calculated by using the formula given below,

1) $C = \frac{T}{R}$ (F) 2) $\varepsilon_r = \frac{Cd}{\varepsilon_0 A}$

RVCE DEPARTMENT EVALUATION OF EXPERIMENT			
Scheme of Evaluation	Course Outcome	Max Marks	
Data Sheet and Experimental Set up (Lab)	CO1	30	476
Conduction of Experiment	CO2	10	10
Tabulation, Calculation & Accuracy	CO3	30	9
Total Marks Scored		50	29
Signature of the Faculty			

Signature of Teacher Incharge

where τ : time constant

ε_r : relative permittivity of the dielectric constant of the dielectric

ε_0 : Absolute permittivity of free space $= 8.854 \times 10^{-12}$

C : Capacitance of the capacitor (F)

R : Resistance (Ω)

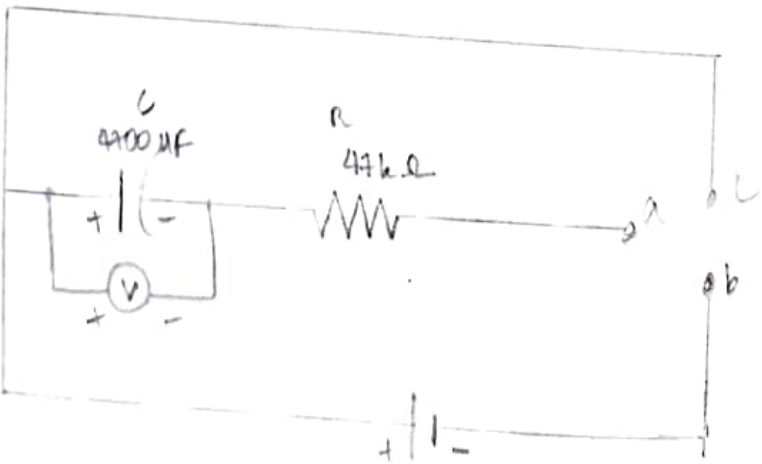
A : Area of each plate (m^2)

d : thickness of the dielectric (m).

Data :

C = 3300 µF	C = 4700 µF
R = 47 kΩ	R = 47 kΩ
L = 47 cm	L = 55 cm
B = 1.5 cm	B = 2.5 cm
d = 80 µm	d = 80 µm

Observation : Circuit diagram.



R = 47 kΩ

Battery Voltage = 2 V

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OBSERVATION / DATA SHEET

Date 12.12.25 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 8

Title DIELECTRIC CONSTANT

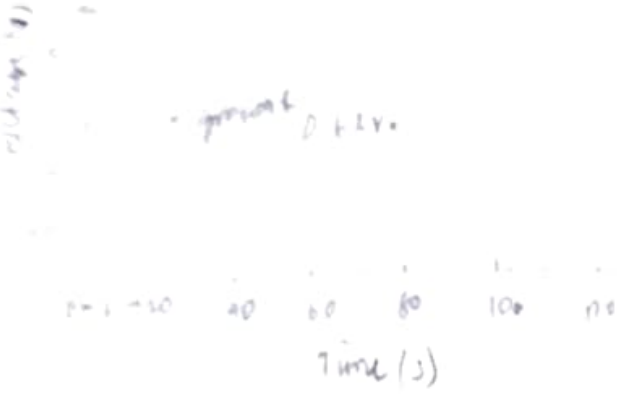
Time in sec (s)	voltage during charging (V)	voltage during discharging (V)	Time in sec (s)	voltage during charging (V)	voltage during discharging (V)
0	0	2.09	690	2.00	0.07
30	0.29	1.78	720	2.01	0.06
60	0.54	1.54	750	2.02	0.05
90	0.74	1.33	780	2.03	0.04
120	0.92	1.15	810	2.04	0.04
150	1.07	0.99	840	2.04	0.03
180	1.20	0.85	870	2.05	0.02
210	1.32	0.74	900	2.06	0.02
240	1.42	0.64	930	2.06	0.02
270	1.52	0.55	960	2.06	0.01
300	1.60	0.48	990	2.07	0.01
330	1.66	0.41	1020	2.07	0.01
360	1.72	0.35	1050	2.07	0.01
390	1.76	0.31	1080	2.08	
420	1.80	0.26	1100	2.08	
450	1.84	0.23	1140	2.08	
480	1.86	0.20	1170	2.09	
510	1.89	0.17	1200	2.09	
540	1.91	0.15	1230	2.09	
570	1.94	0.13	1260	2.09	
600	1.96	0.11	1290		
630	1.98	0.09	1320		
660	1.99	0.08	1350		

12/12

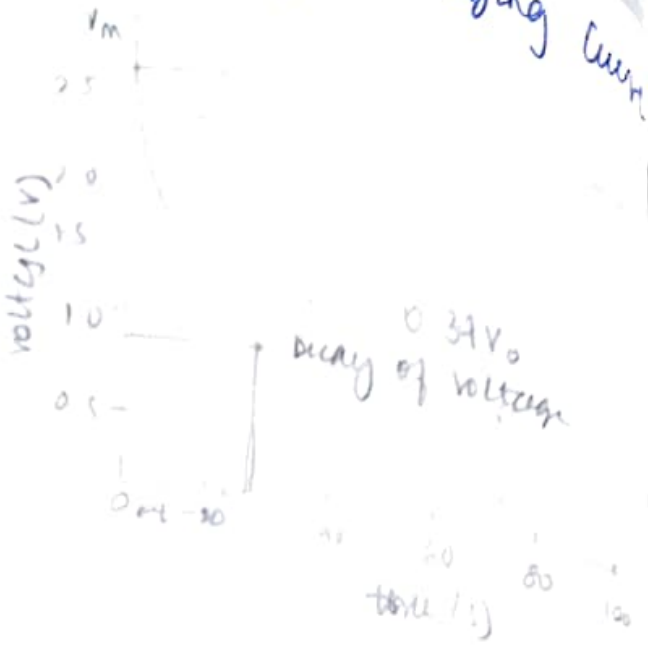
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i) charging curve



ii) Discharging curve



Charge time constant $t_1 = \underline{210}$ s

Discharge time constant $t_2 = \underline{216}$ s

Average time constant $t = \frac{t_1 + t_2}{2} = \underline{213}$ s

Capacitance of +1- capacitor $C = \frac{t}{R} = \underline{4531 \mu F}$

Where R is the resistance and C is the capacitance of the capacitor in the circuit

Dielectric constant is determined by using the formula $\varepsilon_r = \frac{Cd}{\varepsilon_0 A}$

Where t : time constant

ε_r : dielectric constant of the dielectric

ε_0 : Absolute permittivity of free space
$= 8.854 \times 10^{-12}$ F/m

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OBSERVATION / DATA SHEET

Date 12 12 23 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 8

Title Dielectric Constant

C = capacitance of the capacitor (F).

Calculations:

thickness of dielectric medium, d(m)	80×10^{-6}
Area of each plate A (m^2)	137.5×10^{-4}

$C = \frac{I}{\beta} = \frac{213}{47} = 4.531 \times 10^{-3} F$ = 4531 μF

$A = 55 \times 2.5 \times 10^{-4}$

$= 137.5 \times 10^{-4}$

$\varepsilon_r = \frac{Cd}{\varepsilon_0 A}$

$= \frac{4700 \times 10^{-6} \times 80 \times 10^{-6}}{8.854 \times 10^{-12} \times 137.5 \times 10^{-4}} = 3.08$

13/12

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OBSERV

Circuit

R

Time in seconds (s)
0
30
60
90
120
150
180
210
240
270
300
330
360
390
420
450
480
510
540
570
600
630
660
690
720
750
780
810

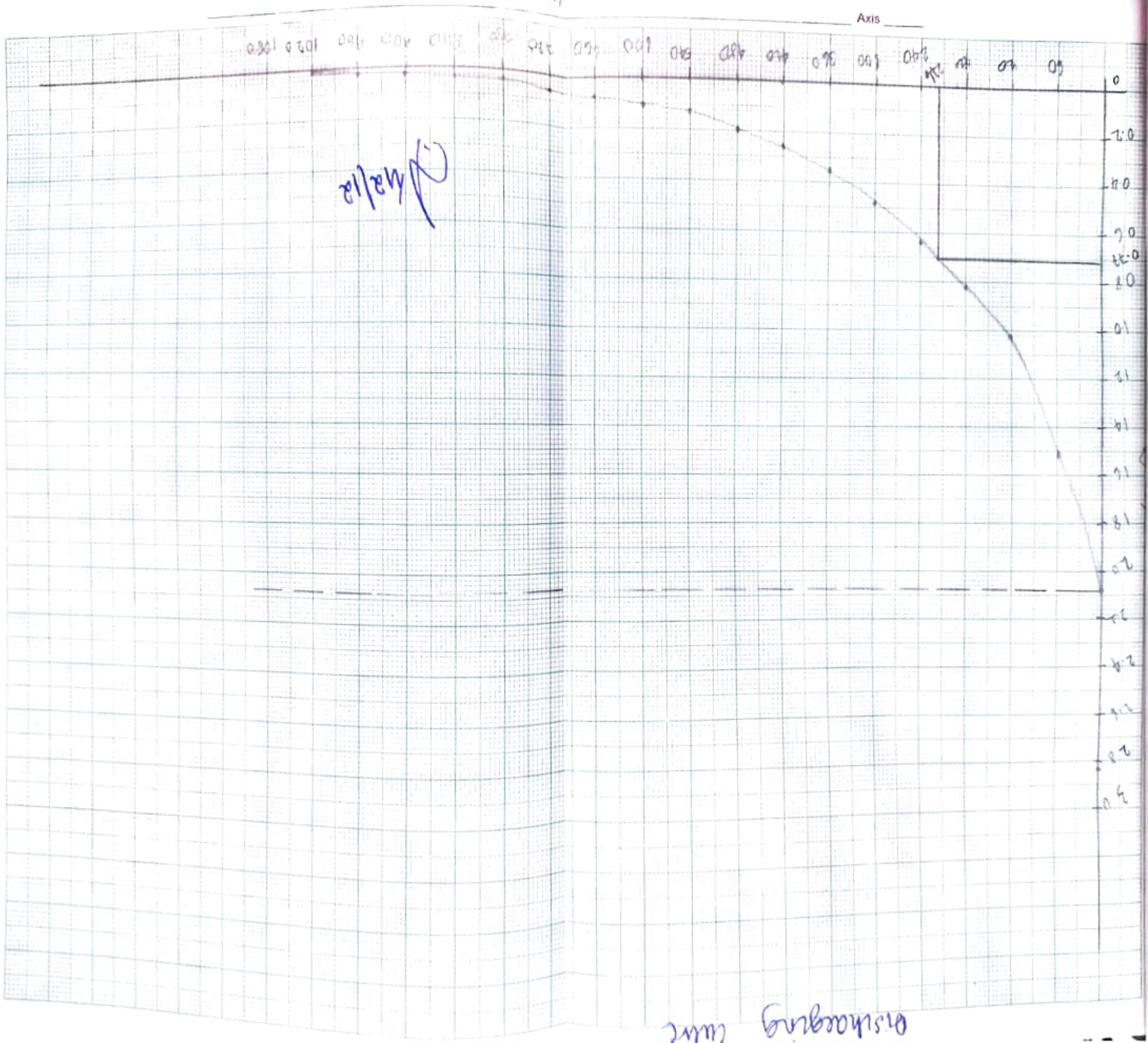
sult : 1) the capacity of parallel plate capacitor

$C = 4531$ F

2) The dielectric constant of the given dielectric material $\varepsilon_r = 3.08$

Date :

Charging curve


Axis
Axis
0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0

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OBSERVATION / DATA SHEET

ate 07.11.2022 Name SHREYA J PATIL

ept./Lab PHYSICS LAB Class _____ Expt./No. 2

le BAND GAP OF A THERMISTOR.

m: To determine the energy gap (E_g) of a thermistor

paratus: Glass Beaker, thermistor, multimeter, thermometer.

ormula: $E_g = \frac{4.606 \times k \times m}{1.6 \times 10^{-19}}$ eV

Where E_g = Energy gap of a given thermistor in eV.

k = Boltzmann constant = 1.381×10^{-23} J/K

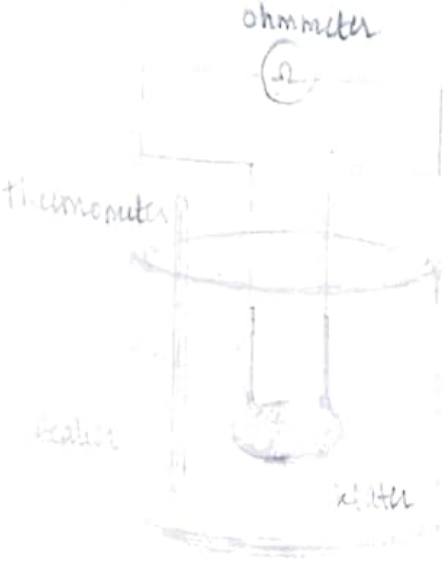
m = slope of the graph of Log R vs $1/T$.

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Scheme of Evaluation	Course Outcome		
Data Sheet and Experimental			6
Conclusion			10
Substitution, Calculation & Accuracy	CO3	10	9
Total Marks Scored		30	29
Signature of the Faculty			

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Diagram:



model Graph:

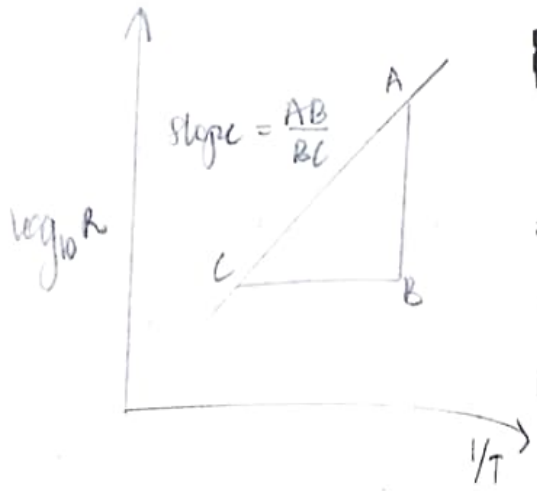


Table:

Sl.no	t Temp (°C)	temp T (k)	R(Ω)	log R	1/T (x 10-3 k-1)
Room temp	26.4		58.6		
1	86	359.15	8.2	0.9138	2.7843
2	84	357.15	8.5	0.9294	2.7999
3	82	355.15	8.8	0.944	2.8157
4	80	353.15	9.2	0.963	2.8316
5	78	351.15	9.6	0.982	2.8477
6	76	349.15	10.1	1.004	2.8640
7	74	347.15	10.6	1.025	2.8805
8	72	345.15	11.1	1.045	2.8972
9	70	343.15	11.7	1.068	2.9141
10.	68	341.15	12.3	1.089	2.9312

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OBSERVATION / DATA SHEET

Date 07.11.2023 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 2

Title Band gap of thermistor.

CALCULATIONS:

$$n = \frac{2.7999 - 2.7843}{0.9299 - 0.9138} = \frac{0.0156}{0.0156} = 0.1$$

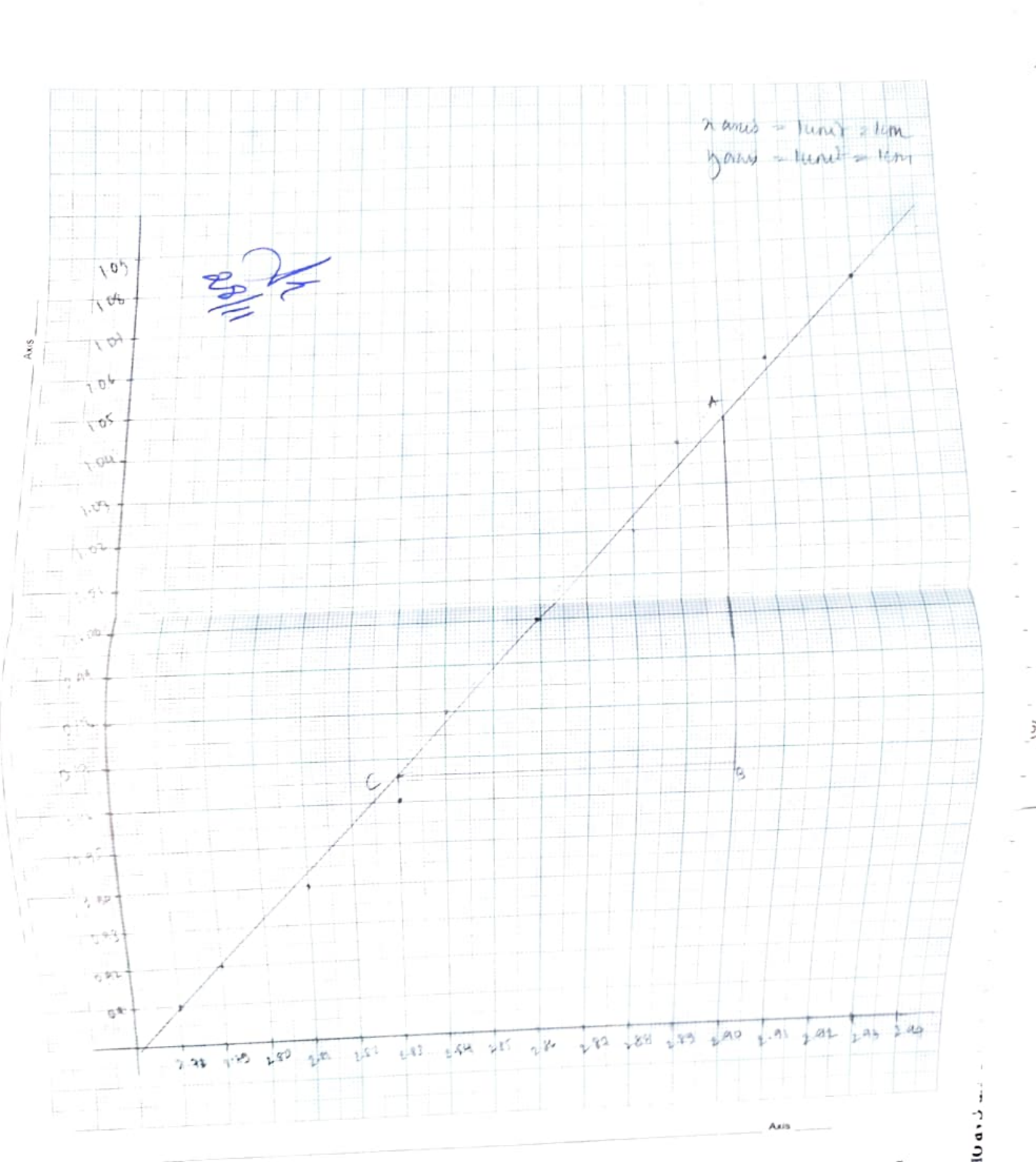
$$n = \frac{1.045 - 0.965}{2.9 - 2.83} = \frac{0.08}{0.07} = 1.142$$

$$E_g = \frac{4.606 \times 1.381 \times 10^{-23} \times 1.42}{1.6 \times 10^{-19}} \quad \frac{6.3608 \times 10^{-23}}{1.6 \times 10^{-19}} = 3.97 \times 10^{-4}$$

$$= \frac{9.0324 \times 10^{-23}}{1.6 \times 10^{-19}} = 0.564 \text{ eV}$$

Result: the energy gap (band gap) of the given thermistor is 0.564 eV.

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x axis = 1unit = 1cm
y axis = 1unit = 1cm
A
B
C
Axis
Axis

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OBSERVATION / DATA SHEET

ate 31 10 2023 Name SHREYA J PATIL

ept./Lab PHYSICS LAB Class _____ Expt./No. 1.

le FERMI ENERGY OF COPPER

AIM: TO determine the Fermi energy of copper

APPARATUS: Multi meter, Beaker, thermometer and copper wire

ORMULA: $E_f = 1.36 \times 10^{-15} \sqrt{\frac{\rho A m}{e}}$ (in J)

$E_f = \frac{1.36 \times 10^{-15} \sqrt{\frac{\rho A m}{e}}}{1.6 \times 10^{-19}}$ = eV

Slope $= m = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$

y intercept $= c = \frac{(\Sigma y)(\Sigma y^2) - (\Sigma x)(\Sigma xy)}{n\Sigma x^2 - (\Sigma x)^2}$

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			4.6
			10
		20	8
		30	28

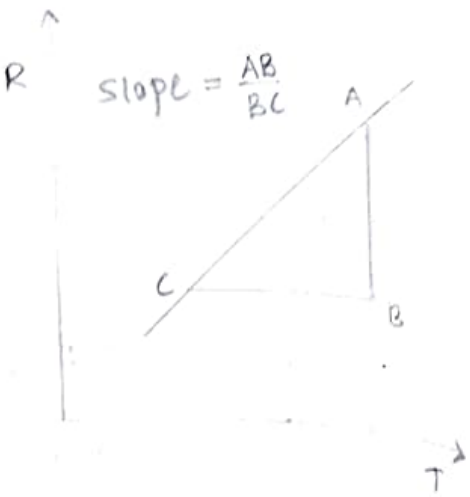
31/10

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Diagram:



model graph:



Tabular Column:

Sl No	Temp (°C) (x)	R(Ω) (y)	x^2	xy
Room Temp	27.9	9.7	778.41	270.63
1	86.8	11.1	7534.24	963.48
2	81.8	11.0	6691.24	899.8
3	76.8	10.8	5898.24	829.44
4	71.8	10.6	5155.24	761.08
5	66.8	10.4	4462.24	694.72
6	61.8	10.3	3819.24	638.54
Sums	$\Sigma x = 473.7$	$\Sigma y = 73.9$	$\Sigma x^2 = 34338.85$	$\Sigma xy = 5055.69$

n	$y = mx + c$
30	9.6505
80	10.8455

$\theta_B =$

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OBSERVATION / DATA SHEET

Date 31.10.2023 Name SHREYA J PATIL

Dept./Lab PHYSICS LAB Class _____ Expt./No. 1

Title Fermi energy of Copper

CALCULATIONS

$$m = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{n\Sigma x^2 - (\Sigma x)^2} = \frac{7 \times 5055.69 - (473.7)(73.9)}{7 \times 34338.85 - (473.7)^2}$$

$$\frac{3589.83 - 3500.43}{240371.95 - 224391.69} = \frac{89.4}{15980.26} = 0.0289$$

$$c = \frac{(\Sigma y)(\Sigma x^2) - (\Sigma x)(\Sigma xy)}{n\Sigma x^2 - (\Sigma x)^2} = \frac{(73.9)(34338.85) - (473.7)(5055.69)}{7 \times 34338.85 - (473.7)^2}$$

$$= \frac{2537641.015 - 2395880.353}{15980.26} = \frac{142760.662}{15980.26} = 8.9335$$

ϕ =

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$E_f = 1.36 \times 10^{-15} \sqrt{\frac{\rho A m}{\ell}}$ (in J)

$\ell = 15\,m$

Radius $= 0.1\,mm$

$\rho = 8960\ kg/m^3$

$A = \pi r^2 = \pi \times \frac{(0.1)}{1000} \times \frac{0.1}{1000} = \pi \times 10^{-8}$

E_f (in J) $= 1.36 \times 10^{-15} \sqrt{\frac{8960 \times \pi \times 10^{-8} \times 0.0289}{15}} = 1.36 \times 10^{-15} \sqrt{\frac{8.130 \times 10^{-4}}{15}}$

$= 1.36 \times 10^{-15} \sqrt{5.420 \times 10^{-7}} = 1.36 \times 10^{-15} \times 7.362 \times 10^{-4}$

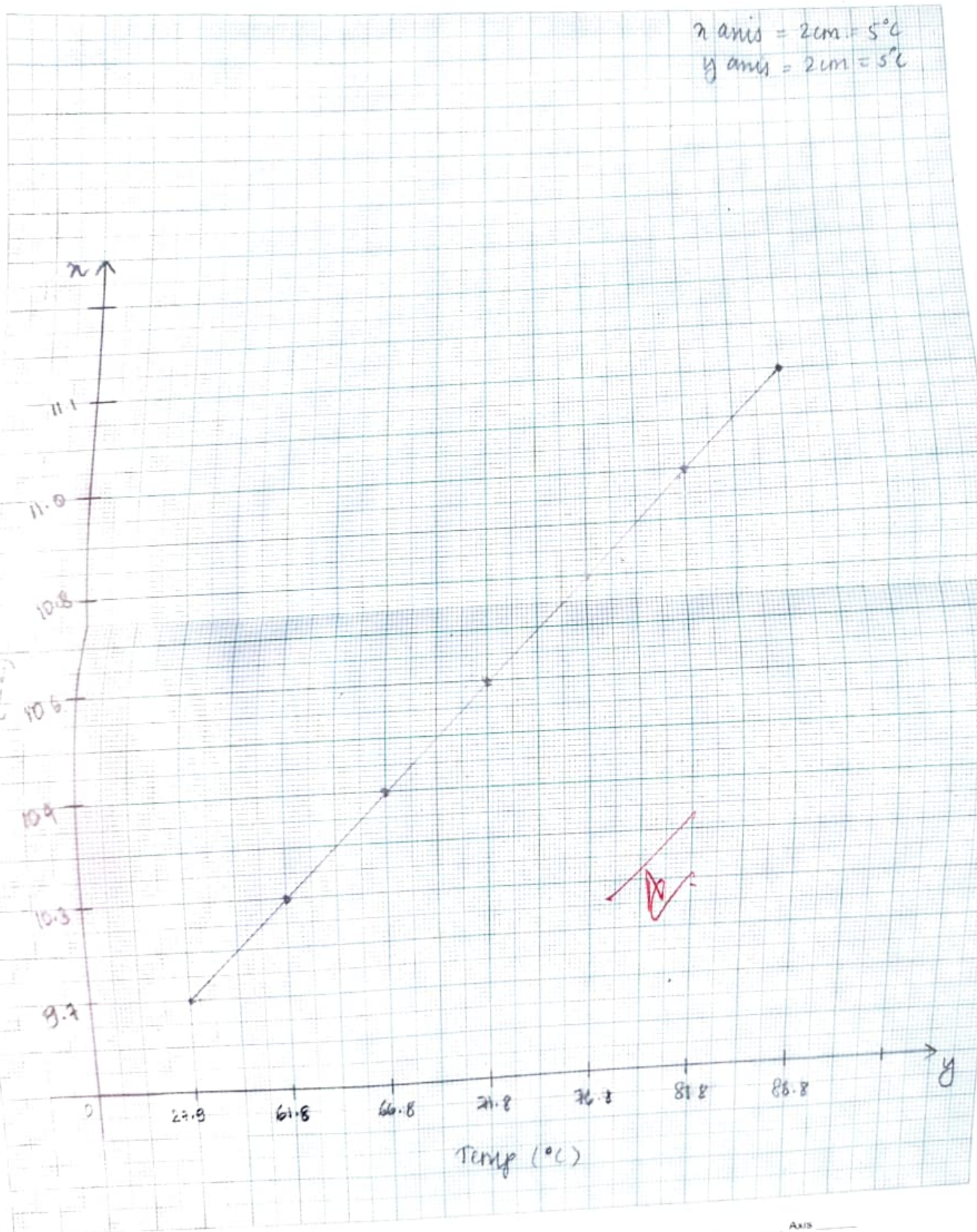
$= 1.001 \times 10^{-18}$ J

E_f (in eV) $= \frac{1.001 \times 10^{-18}}{1.6 \times 10^{-19}} = 6.25$ eV

RESULT : the Fermi energy of copper $E_f = \underline{1.001 \times 10^{-18}}$ J

$E_f = \underline{6.25}$ eV

x axis = 2cm = 5°C
y axis = 2cm = 5°C



Axis

Axis

FERMI ENE